



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

Subject: Programming with Java

ASSIGNMENT No :2

TITLE :Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Name : Atharva Ambilwade

PRN : 25070122055

Class : CSE

Batch : 2025-29

Assignment 2: Methods and Static Members

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Theory

Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

Constructors make object creation easy and efficient by assigning initial values to data members. They help avoid writing separate methods for initialization and ensure that every object starts with the required information. Different types of constructors provide flexibility in creating objects with default values, user-defined values, or copied values. As a result, constructors make Java programs more organized, readable, and easier to maintain.



Code and Output Screenshots

- Default Constructor

```
J Student.java > Language Support for Java(TM) by Red Hat > Student
1  public class Student {
2      String name;
3      int age;
4
5      // Default Constructor
6      Student() {
7          name = "Atharva";
8          age = 18;
9      }
10
11     void display() {
12         System.out.println("Name: " + name);
13         System.out.println("Age: " + age);
14     }
15
16     Run main | Debug main | Run | Debug
17     public static void main(String[] args) {
18         Student s1 = new Student();
19         s1.display();
20     }
}
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\LENOVO\Documents\JAVA Assignments> javac Student.java
● PS C:\Users\LENOVO\Documents\JAVA Assignments> java Student
Name: Atharva
Age: 18
❖ PS C:\Users\LENOVO\Documents\JAVA Assignments> |
```



- Parameterized Constructor

J Student.java > Java > Student

```
1  class Student {
2      String name;
3      int age;
4
5      // Parameterized Constructor
6      Student(String n, int a) {
7          name = n;
8          age = a;
9      }
10
11     void display() {
12         System.out.println("Name: " + name);
13         System.out.println("Age: " + age);
14     }
15
16     Run main | Debug main | Run | Debug
17     public static void main(String[] args) {
18         Student s1 = new Student("Rajneesh", 20);
19         s1.display();
20     }
21 }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\LENOVO\Documents\JAVA Assignments> javac Student.java
● PS C:\Users\LENOVO\Documents\JAVA Assignments> java Student
Name: Rajneesh
Age: 20
❖ PS C:\Users\LENOVO\Documents\JAVA Assignments> 
```



- Copy Constructor

```
J Student.java > Java > Student
1  class Student {
2      String name;
3      int age;
4
5      // Parameterized Constructor
6      Student(String n, int a) {
7          name = n;
8          age = a;
9      }
10
11     // Copy Constructor
12     Student(Student s) {
13         name = s.name;
14         age = s.age;
15     }
16
17     void display() {
18         System.out.println("Name: " + name);
19         System.out.println("Age: " + age);
20     }
21
22     Run main | Debug main | Run | Debug
23     public static void main(String[] args) {
24         Student s1 = new Student("Bhola", 21);
25         Student s2 = new Student(s1);
26
27         s1.display();
28         s2.display();
29     }
}
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\LENOVO\Documents\JAVA Assignments> javac Student.java
● PS C:\Users\LENOVO\Documents\JAVA Assignments> java Student
Name: Bhola
Age: 21
Name: Bhola
Age: 21
❖ PS C:\Users\LENOVO\Documents\JAVA Assignments> 
```



Conclusion

This assignment demonstrated the use of default, parameterized, and copy constructors in Java for object initialization. It helped in understanding how constructors simplify object creation, improve code organization, and support effective data management in object-oriented programming.

Exercise

1. Create a Student class using default and parameterized constructors to initialize student name and roll number.

```
J Student.java > Java > Student
1  class Student {
2      String name;
3      int rollNo;
4
5      // Default Constructor
6      Student() {
7          name = "Atharva";
8          rollNo = 055;
9      }
10
11     // Parameterized Constructor
12     Student(String n, int r) {
13         name = n;
14         rollNo = r;
15     }
16
17     void display() {
18         System.out.println("Name: " + name);
19         System.out.println("Roll No: " + rollNo);
20     }
21
22     Run main | Debug main | Run | Debug
23     public static void main(String[] args) {
24         // Object using default constructor
25         Student s1 = new Student();
26
27         // Object using parameterized constructor
28         Student s2 = new Student("Avdhoot", 056);
29
30         System.out.println("Default Constructor:");
31         s1.display();
32
33         System.out.println();
34
35         System.out.println("Parameterized Constructor:");
36         s2.display();
37     }
38 }
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\LENOVO\Documents\JAVA Assignments> java Student
Name: Atharva
Roll No: 45

Parameterized Constructor:
Name: Avdhoot
Roll No: 46
PS C:\Users\LENOVO\Documents\JAVA Assignments> |
```



2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

```
J Mobile.java > Java > Mobile
1  class Mobile {
2      String brand;
3      int price;
4
5      // Default Constructor
6      Mobile() {
7          brand = "Samsung";
8          price = 20000;
9      }
10
11     // Parameterized Constructor
12     Mobile(String b, int p) {
13         brand = b;
14         price = p;
15     }
16
17     // Copy Constructor
18     Mobile(Mobile m) {
19         brand = m.brand;
20         price = m.price;
21     }
22
23     void display() {
24         System.out.println(brand + " - " + price + " rs");
25     }
26
27     Run main | Debug main | Run | Debug
28     public static void main(String[] args) {
29         Mobile m1 = new Mobile();
30         Mobile m2 = new Mobile("Apple", 80000);
31         Mobile m3 = new Mobile(m2);
32
33         m1.display();
34         m2.display();
35         m3.display();
36     }

```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\LENOVO\Documents\JAVA Assignments> javac Mobile.java
PS C:\Users\LENOVO\Documents\JAVA Assignments> java Mobile
Samsung - 20000 rs
Apple - 80000 rs
Apple - 80000 rs
PS C:\Users\LENOVO\Documents\JAVA Assignments>
```